

```
sentence1 = ""I will walk 500 miles and I would walk 500 more. Just to be the
a thousand miles to fall down at your door!""
```

```
sentence2 = "I played the play playfully as the players were playing in the"
```

```
from nltk import word_tokenize, sent_tokenize
```

```
import nltk
nltk.download('punkt_tab')
print("-----Tokenized Words-----")
print('Tokenized words:', word_tokenize(sentence1))
print("\n")
```

```
-----Tokenized Words-----
Tokenized words: ['I', 'will', 'walk', '500', 'miles', 'and', 'I', 'would', 'walk', '500', 'more', '.', 'Just', 'to', 'be', 'the
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
```

```
print("-----Tokenized Sentences-----")
print('Tokenized sentences:', sent_tokenize(sentence1))
print("\n")
```

```
-----Tokenized Sentences-----
Tokenized sentences: ['I will walk 500 miles and I would walk 500 more.', 'Just to be the\na thousand miles to fall down at your
```

```
import nltk
nltk.download('averaged_perceptron_tagger_eng')
from nltk import pos_tag
print("-----POS Tagging-----")
token = word_tokenize(sentence1) + word_tokenize(sentence2)
print('POS tagged:', pos_tag(token))
print("\n")
```

```
-----POS Tagging-----
POS tagged: [('I', 'PRP'), ('will', 'MD'), ('walk', 'VB'), ('500', 'CD'), ('miles', 'NNS'), ('and', 'CC'), ('I', 'PRP'), ('would
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!
```

```
print("-----Stop-Words Removal-----")
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
stop_words = set(stopwords.words('english'))
```

```
token = word_tokenize(sentence1)
cleaned_token = []
for word in token:
    if word not in stop_words:
        cleaned_token.append(word)
print("Unclean version:", token)
print("\n")
print("Cleaned version:", cleaned_token)
print("\n")
```

```
-----Stop-Words Removal-----
Unclean version: ['I', 'will', 'walk', '500', 'miles', 'and', 'I', 'would', 'walk', '500', 'more', '.', 'Just', 'to', 'be', 'the
```

```
Cleaned version: ['I', 'walk', '500', 'miles', 'I', 'would', 'walk', '500', '.', 'Just', 'thousand', 'miles', 'fall', 'door', '!
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
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```
print("-----Stemming-----")
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
token = word_tokenize(sentence2)
stemmed = [stemmer.stem(word) for word in token]
print("Stemmed words:", stemmed)
print("\n")
```

```
-----Stemming-----
Stemmed words: ['i', 'play', 'the', 'play', 'play', 'as', 'the', 'player', 'were', 'play', 'in', 'the']
```

```
print("-----Lemmatization-----")
import nltk
nltk.download('wordnet')
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
token = word_tokenize(sentence2)
lemmatized = [lemmatizer.lemmatize(word) for word in token]
print("Lemmatized words:", lemmatized)
print("\n")
```

```
-----Lemmatization-----
Lemmatized words: ['I', 'played', 'the', 'play', 'playfully', 'a', 'the', 'player', 'were', 'playing', 'in', 'the']
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
```